

Sources of Innovation

Chapter 6 — Managing Innovation

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Learning Objectives

By the end of this chapter, you will be able to:

- **Identify** the diverse sources of innovation beyond R&D.
- **Distinguish** between "Technology Push" and "Market Pull."
- **Analyze** the role of Users as active innovators (User Innovation).
- **Evaluate** the potential of Crowdsourcing and Open Innovation.
- **Understand** how accidents and shocks can trigger innovation.

The Innovation Map

Innovation is rarely a solitary act of genius. It comes from many directions.

The Myth of the Lab

We often assume innovation only comes from "white coats" in R&D labs. In reality, sources are distributed widely across the internal and external environment.

Key Sources:

- Knowledge Push (R&D)
- Need Pull (Market)
- Users
- Accidents
- Regulation

Knowledge Push vs. Need Pull

Knowledge Push



Need Pull



Reality: Most successful innovation involves a coupling of both.

"Necessity is the mother of invention."

Shocks/Crisis:

- Wars, pandemics, or resource shortages force rapid innovation.
- Example: Synthetic rubber during WWII; mRNA vaccines during COVID-19.

Regulation:

- Often seen as a constraint, but acts as a powerful stimulus.
- Example: Environmental regulations driving Electric Vehicle (EV) innovation.

The User as Innovator (Von Hippel)

Users are not just passive consumers. They often innovate to solve their own problems.

Lead Users

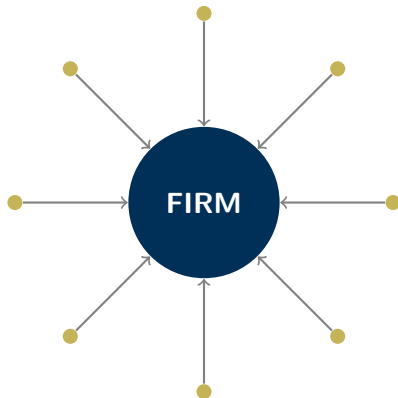
Users whose needs are ahead of the general market. They develop solutions for themselves because no commercial product exists yet.

Examples:

- Mountain Bikes (invented by enthusiasts, not bike companies).
- Open Source Software (Linux).
- Medical devices (Surgeons modifying tools).

Crowdsourcing: Wisdom of the Crowd

Using the internet to outsource tasks to a distributed network of people.



The Crowd submits ideas, solutions, or designs.

Serendipity

Accidental discoveries. But as Pasteur said, "Chance favors the prepared mind."

- Example: Penicillin, Post-it Notes, Viagra.

Recombinant Innovation:

- Innovation often comes from crossing ideas from different worlds.
- Example: Ford's assembly line came from observing meat-packing plants.
- *Crossover*: Transferring a solution from Sector A to Sector B.

Watching Others: Benchmarking & Imitation

Innovation doesn't always mean "first to the world." It can mean "new to us."

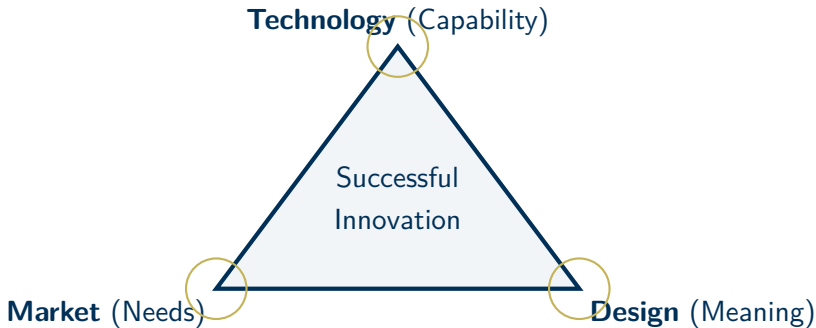
- **Reverse Engineering:** Taking apart competitor products to understand them.
- **Benchmarking:** Comparing your processes against "best in class" (even outside your industry).
- **Fast Follower Strategy:** Letting pioneers take the risk, then entering the market with a better/cheaper version.

Creative Imitation

Copying the substance of an innovation but adapting it to a new context or market.

Design-Driven Innovation

Innovation is not just about functionality (technology) or utility (market). It is also about **Meaning**.



Example: Apple iPod (Tech) + iTunes (Market) + White Earbuds/Cool Factor (Meaning).

How do we find sources from the future?

- **Delphi Method:** Aggregating expert opinions anonymously to predict timelines.
- **Scenario Planning:** creating multiple "stories" about the future (Best case, Worst case, Wildcards) to test resilience.
- **Trend Extrapolation:** Projecting current data forward (risky if trends break).

Goal

Not to predict the future perfectly, but to prepare the organization for multiple possibilities.

Summary:

1. Innovation sources are ubiquitous: R&D, Customers, Users, Competitors, and Accidents.
2. **Users** (Lead Users) are increasingly important sources of new product concepts.
3. **Open Innovation** (Crowdsourcing) allows firms to tap into global brainpower.
4. Successful firms build "Search Strategies" to monitor all these channels simultaneously.

Thank You

Questions?